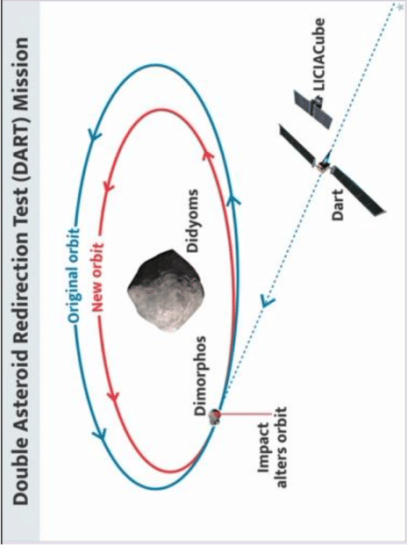


DART MISSION	● Objective→deflect an asteroid to change its motion in space and save earth from it's impact ● DART's demo conducted on asteroid system Didymos and moonlet  Double Asteroid Redirection Test (DART) Mission
SOFIA Mission Stratospheric Observatory for Infrared Astronomy	● SOFIA is an infrared telescope ○ Collaboration between NASA and German Space Agency ● Objective→ ○ understanding star birth and death ○ Formation of new solar systems ● Discovery ○ Discovered helium hydride—>First molecule formed in universe almost 14 billion years ago
ISRO	



Hybrid propulsion system	- Unlike solid-solid or liquid-liquid combinations, a hybrid motor uses solid fuel and a liquid oxidizer. - The motor used Hydroxyl-terminated polybutadiene (HTPB) as fuel and liquid oxygen (LOX) as the oxidizer. Benefits: more efficient, greener and safer to handle.
HS200 booster for Gaganyaan Programme	- It is the world's second largest operational booster using solid propellants Gaganyaan's objective - indigenous capability to undertake human spaceflight to Low Earth Orbit (an orbit of 2,000km or less).
Inflatable Aerodynamic Decelerator (IAD)	ISRO has successfully tested IAD technology to land missions on Venus & Mars. - Used for decelerating an object descending through the atmosphere. - IAD is made of Kevlar fabric coated with polychloroprene. - Kevlar has properties like high tensile strength, toughness, thermal stability etc.



Rohini Sounding Rockets	• ISRO launched Rohini RH-200 sounding rocket. • Sounding rockets are ○ One or two stage solid propellant rockets ○ Used for probing the upper atmospheric regions and for space research. • First sounding rocket to be launched from India was American Nike-Apache in 1963.
RISAT-2 satellite	• RISAT 2 is a radar-imaging satellite • The RISAT satellites are equipped with a synthetic aperture radar (SAR) • Applications ○ It can take pictures of the earth during day and night, and also under cloudy conditions. ○ It helps round-the-clock border surveillance along with checking infiltration and keeping an eye on terror or anti-national activities across the borders.



Aditya L1 mission	• Objective→ study of Sun's corona, solar emissions, solar winds and flares, and Coronal Mass Ejections (CMEs) • Other solar missions: ◦ European Space Agency's Solar Orbiter, ◦ NASA's Parker Solar Probe.
Space Tech Innovation Network (SpIN)	• ISRO has signed an MoU with Social Alpha to launch SpIN. • SpIN is India's first dedicated platform space entrepreneurial ecosystem.
SPACE PHENOMENON AND EXPERIMENTS	
Solar Eclipse	• A solar eclipse occurs when the Moon gets between Earth and Sun, and the moon casts a shadow over Earth. • Solar eclipses happen only at the new moon phase. • Different types of solar eclipses are: ◦ Total solar eclipse: When moon



	passes between Sun and Earth, completely blocking the face of Sun. ○ Partial solar eclipse: When sun, moon and Earth are not exactly lined up. Only a part of the Sun will appear to be covered, giving it a crescent shape. ○ Annular solar eclipse: When the moon is farthest from Earth. It does not block the entire view of the sun. This looks like a ring around the moon. ○ Hybrid solar eclipse: Because Earth's surface is curved, sometimes an eclipse can shift between annular and total as the Moon's shadow moves across the globe.
Blood Moon	● Blood Moon, commonly known as total lunar eclipse, occurs when moon passes through darkest part of Earth's shadow, known as the umbra. ○ It is called blood moon because of reddish hue. ● Other Types of Lunar Eclipse: ○ Partial Lunar Eclipse: When only a part of moon enters earth's shadow.



	○ Penumbral Lunar Eclipse: When the moon enters the Earth's penumbra.
Geomagnetic Storm	● GMS is a disturbance in the earth's magnetosphere, which is the area around the planet controlled by its magnetic field. ○ When Coronal Mass Ejections (CME) collide with the Earth, it causes GMS. ○ CME is a large expulsion of plasma and magnetic field from the sun's corona, or upper atmosphere. ● Impact of geomagnetic storms—>PYQ ○ Disrupt high-frequency radio broadcasts and global positioning system (GPS) devices, damage satellite electronics, can affect power supply on earth.
Coronal Holes	● Coronal holes are regions on the sun's surface from where fast solar wind gushes out into space. ● These coronal holes can cause a solar storm on Earth as they release a complex stream of solar winds.
Sunspots	● Sunspots are areas that appear dark on the surface of the Sun.



	○ They appear dark because they are cooler than other parts of the Sun's surface. ○ They form at areas where magnetic fields are particularly strong thus keeping some of the heat within Sun from reaching the surface. ● Magnetic field lines near sunspots can cause a sudden explosion of energy called a solar flare.
Fast Radio Bursts (FRBs)	● FRBs are bright bursts of radio waves whose durations lie in the millisecond-scale, because of which it is difficult to detect them and determine their position in the sky. ○ Produced by astronomical objects with changing magnetic fields
Gamma Ray Bursts (GRBs)	● GRBs are flashes of high-energy radiation arising from energetic cosmic explosions. ○ GRBs are the most powerful explosions the universe has seen since the Big Bang. ● Astronomers recorded binary merger emitting long GRB twinned with a kilonova emissions.



	○ Kilonova occurs when two compact objects, like binary neutron stars or a neutron star and a black hole, collide.
Betelgeuse	● It is the bright red supergiant on the shoulder of Orion. ○ In 2019 Betelgeuse likely underwent an enormous surface mass ejection (SME) ○ It ejected 400 billion times more mass than a typical event on other stars. ○ SME happens when a star expels large amounts of plasma and magnetic flux into the surrounding space. ● A red giant is a dying star in the final stages of stellar evolution.
White Dwarf	● White dwarfs are stars that have burned up all of hydrogen they once used as nuclear fuel.



Plants in Moon soil	- Scientists have for the first time successfully grown plants in the Moon's soil. - This lunar soil, also called regolith, was brought to Earth from the Moon by the Apollo-era astronauts.
SPACE OBJECTS	
New asteroid 2022 AP7 discovered	- Astronomers discovered a new Potentially Hazardous Asteroid named 2022 AP7. - It was 1.5 kilometer wide. Any asteroid over 1km in size is considered a planet killer. - It will cross Earth's orbit
Exoplanets	Detection of barium in the upper atmosphere of two giant exoplanet (WASP- 76b and WASP-121b) for the first time. WASP-76b and WASP-121b are Ultra-hot Jupiters, a class of hot gaseous planets that matches the size of Jupiter.
Bernardinelli-Bernstein comet	largest icy comet nucleus ever seen by astronomers. - BBC - Originated in the Oort cloud, a distant region of the solar system that is predicted to be the source of most comets.
MISC	



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Dark SKY Reserve	● India's first dark sky reserve at Hanle in Ladakh as a part of Changthang Wildlife Sanctuary. ○ World's highest-located sites for optical, infra-red, and gamma-ray telescopes. ● Dark Sky Reserve ○ Place with policies to ensure minimal artificial light interference. ○ International Dark Sky Association, a U.S.-based non-profit, designates places as International Dark Sky Places
Sampurnanand Optical Telescope (SOT)	● Located at Nainital ● SOT has been used for optical observations of comets, occultation by planets and asteroids, star forming regions and star clusters, active galactic nuclei, etc. ● Important discovery made: ○ Discovery of rings of Neptune; ○ Contributed to detection of rings around Uranus and additional rings around Saturn; ○ First detection of optical afterglows of Gamma-ray-bursts (GRBs); ○ Microlensing event; ○ Discovery of quakes in various stars (under Nainital-Cape Survey program) etc.
Shaped Antenna measurement of	● SARAS 3 is a precision radio telescope to detect extremely faint radio wave signals from Cosmic Dawn.



the background Radio Spectrum (SARAS 3) Telescope	- It is indigenously designed and built at Raman Research Institute - Located at Dandiganahalli Lake and Sharavati backwaters, in Karnataka. - SARAS 3 has helped determine properties of the earliest radio luminous galaxies formed 200 million years after the Big Bang, a period known as the Cosmic Dawn.
Liquid-Mirror Telescope (LMT)	India's first and Asia's largest liquid-mirror telescope - commissioned at the Devasthal Observatory of Aryabhata Research Institute of Observational Sciences (ARIES) in Nainital (Uttarakhand). - LMT will observe asteroids, supernovae, space debris and all other celestial objects. Rotating mirror made up of a thin film of liquid mercury to collect and focus light. - Primary mirror is liquid It cannot be turned and pointed in any direction It watches the sky as the Earth rotates. - Built by astronomers from India, Belgium and Canada
Space Bricks	Bricks of complex shapes from Martian soil with the help of bacteria and urea. - Made by ISRO and IISc - Made by mixing Martian soil with guar gum, a bacterium called Sporosarcina pasteurii, urea and nickel chloride (NiCl₂).
Tiangong	- Space station built by China 2nd such body in low-Earth orbit after the NASA- led International Space Station (ISS). - Tiangong will be much smaller and lighter than ISS.

